Project Proposal

Level 04

Domain-Generalised Detection of Small and Thin Road Damage in Unseen Countries: Size-Aware Label Assignment, Scale-Consistency Learning and Missed-Detection Prediction

Group Ascend

Faculty of Information Technology

University of Moratuwa

2026

Project Proposal

Level 04

Domain-Generalised Detection of Small and Thin Road Damage in Unseen Countries: Size-Aware Label Assignment, Scale-Consistency Learning and Missed-Detection Prediction

Group Ascend

225019D Chamikara J.R.H.

225007N Amri B.M.M.

225040H Imasha G.G.P.

Supervisors

Prof. (Mrs) Thanuja Sandanayake

Dr. A.L.A. Romesh R. Thanuja

Faculty of Information Technology

University of Moratuwa

2026

Contents

1. Introduction 1

2. Background & Motivation 1

3. Problem in Brief: Missed Small Damage on Unseen Countries 1

4. Aim & Objectives 1

5. Source-Only Learning and Monitoring for Small Road Damage 2

6. Resource Requirements 3

Reference 3

Appendix A: Plan of Action 4

1. Introduction

Road agencies increasingly survey pavements with low-cost smartphone and dashboard cameras, and deep-learning object detectors can locate cracks and potholes in these images automatically [1]. A detector trained on one country's roads, however, loses much of its accuracy in another country: a detector trained on Japanese roads reached only 0.20 mAP@0.5 on Indian roads [7], and a YOLOv8s model trained in India reached 0.32 mAP@0.5 in South Africa [2]. Countries such as Sri Lanka have no large public labelled road-damage dataset that we could find, so a detector used there would have to be trained on other countries' images. This project studies, on the public multi-country RDD2022 dataset [1], how to make a YOLO11 detector [6] work better on a country it has never seen, without using any images from that country, with a focus on small and thin damage such as hairline cracks. Three members each test one method and compare it against a matched alternative.

2. Background & Motivation

2.1 Cross-country road damage detection

RDD2022 contains 47,420 road images from six countries, with cracks and potholes labelled as boxes [1]. Detectors trained on some of these countries degrade on others because pavements, road markings, cameras, lighting and labelling styles differ [2], [7]. Three remedies exist. Fine-tuning on labelled images from the new country restores accuracy but needs local labels [2]. Unsupervised domain adaptation, such as DA-RDD, uses unlabelled images from the new country and aligns features across countries [7]. Source-only domain generalisation uses no target-country data at all; for road damage it was introduced by MAMDG, which aligns features across several source countries with domain discriminators and a triplet loss [3]. The source-only setting is the realistic one for a country with no collected data, and it is the setting of this project.

2.2 Why small and thin damage

Hairline cracks occupy only a few pixels and are the damage detectors miss first. In modern one-stage detectors, a task-aligned assigner decides which predictions are trained for each labelled box using an overlap score [5]; for a box a few pixels wide, a two-pixel offset removes about half of the overlap, so thin damage may be weakly supervised. A normalised Wasserstein distance (NWD) was proposed to measure tiny boxes more gently [10]. Agreement between predictions on differently transformed views improves generalisation to unseen domains [4], and teacher models averaged over training steps give stable targets for such agreement [9]. Finally, a missed object produces no output at all; failure-prediction models that read a detector's internal features have flagged missed traffic signs [8]. To our knowledge, none of these has been tested for road-damage detection on an unseen country without target data.

2.3 Motivation and capability

A detector that transfers to new countries, and that indicates where it has probably missed damage, would let agencies without labelled data survey roads with ordinary cameras and review only flagged areas. The team has Python, machine learning and software development experience, including industry software work, and will build a shared baseline together before the individual studies begin.

3. Problem in Brief: Missed Small Damage on Unseen Countries

When a road-damage detector trained on some countries is applied to a country whose images were never used for training, small and thin damage is detected unreliably, and the user receives no indication of what was missed. Existing remedies either require data from the target country [2], [7] or change the shared feature representation [3]. Whether changing how thin damage is supervised, how small damage is trained across scales, and how missed damage is monitored helps in the source-only setting has not, to our knowledge, been tested.

4. Aim & Objectives

Aim: The aim of this project is to improve and monitor the detection of small and thin road damage on countries unseen during training, using source-only training of a YOLO11 detector on the RDD2022 dataset.

Objectives:

- Study cross-country road damage detection and the literature on tiny-object assignment, consistency training and detector failure prediction.
- Build a leave-one-country-out benchmark on RDD2022 and measure the accuracy gap between same-country and unseen-country testing for the baseline detector.
- Design and implement three source-only methods (Section 5.2), one per member.
- Evaluate each method against a matched alternative on held-out countries, with comparisons and measures fixed in advance.
- Build a demonstration web application and prepare the final documentation.

5. Source-Only Learning and Monitoring for Small Road Damage

5.1 Shared experimental framework

The study uses the labelled RDD2022 images from Japan, India, the United States and the Czech Republic (25,846 images) [1]. In each of four folds, the detector is trained on three countries and tested on the fourth, which is never used for training or tuning; one source country acts as an inner validation country for all development decisions. Duplicate images are kept within one split. The shared baseline is YOLO11s with COCO-pretrained weights at 640 px, using Ultralytics release v8.4.192 [6]; a strong data-augmentation detector serves as a reference. A week-6 pilot measures training time and the baseline's same-country versus unseen-country accuracy gap before the member studies start. Figure 1 shows how the three member studies share this framework.

RDD2022: Japan, India, US, Czech Republic Train on 3 countries, test on the unseen 4th	**Shared YOLO11s baseline** COCO weights, 640 px, Ultralytics v8.4.192	**Week-6 pilot** Same-country vs unseen-country gap
A: Label assignment Short-side-normalised NWD in the assigner	**B: Scale consistency** Teacher–student agreement on small/thin boxes	**C: Missed-detection prediction** 40 × 40 miss map from frozen features

Figure 1: Shared framework and the three member studies

5.2 Member studies

Each member changes one part of the detection pipeline and is judged against a matched alternative on the held-out countries (Table 1).

Table 1: Member studies, comparisons and primary measures

Member	Method	Principal comparison	Primary measure
A	Replace the overlap score in YOLO11's task-aligned assigner [5] with NWD [10] divided by each labelled box's short side	Plain NWD with one fixed constant, equal candidate selection and tuning	Held-out mAP50:95, overall and for small or thin boxes
B	Scale-consistency loss: an averaged teacher [9] sees the full image, the student a half-size copy; agreement enforced only on small and thin boxes [4]	Same loss on all boxes; and on a random set of boxes of equal size	Held-out recall of small or thin damage, with false positives per image

Member	Method	Principal comparison	Primary measure
C	Small convolutional head on the frozen detector's features and low-confidence detections, predicting a 40 × 40 grid of missed small or thin boxes [8]	Same head trained on the low-confidence detection map only	Held-out cell-level AUPRC for missed damage

Member A tests whether judging prediction error relative to how thin a box is changes which predictions are trained on thin damage and improves transfer; assignment counts and training weights are logged by box size. **Member B** tests whether agreement training focused on small and thin damage improves their recall more than the same training applied generically. **Member C** tests whether the detector's internal features predict where small or thin damage was missed, beyond its own low-confidence outputs; the warnings flag areas for review and do not recover the missed damage.

5.3 Evaluation

Results are reported per held-out country, as a four-country mean and for the worst country. Uncertainty is estimated with a paired bootstrap over whole images, and principal comparisons use three training seeds. Code, settings and analysis are frozen before the held-out countries are scored once. Results are benchmark evidence only; no claim is made about savings, road safety or performance in Sri Lanka.

5.4 Prototype and feasibility

A web application (FastAPI service with an exported ONNX model and a React page) accepts a road image (input), runs the detector and Member C's monitor (process), and shows damage boxes, calibrated scores and a missed-damage heat map (output) for a road inspector (user). Feasibility rests on public data, open-source software and modest GPU time: the default study needs about 72–84 final detector trainings plus about 20 screened development runs, and Member C's heads train on cached features in a few GPU-hours. The week-6 pilot sets the final scope against the GPU hours actually available, following a pre-agreed order of cuts. The plan of action is given in Appendix A.

6. Resource Requirements

- GPU: one NVIDIA GPU with at least 12 GB memory, about 15–25 hours per week, using Kaggle and Google Colab GPU sessions.
- Personal computers with at least 16 GB RAM for development and analysis.
- Storage: about 100 GB for the dataset, cached features and model checkpoints.
- Software: Python 3.10+, PyTorch, Ultralytics v8.4.192, scikit-learn, FastAPI, React; all free and open source.
- Data: RDD2022, publicly available under CC BY-SA 4.0 [1]; no new data collection.
- Version control and collaboration: Git and GitHub.

Reference

[1] Arya, D., Maeda, H., Ghosh, S. K., Toshniwal, D. & Sekimoto, Y. (2024), RDD2022: A multi-national image dataset for automatic road damage detection, Geoscience Data Journal, doi:10.1002/gdj3.260

[2] Babawarun, T., Pandelani, T. & Ngwangwa, H. (2025), Investigation of the transferability of measured data for application of YOLOv8s in the identification of road defects: an SA–Indian case study, Sustainability, 17(23), 10641

[3] Chen, Z., Zhang, J., Lin, Y., Wu, R., Zhou, H. & Wang, W. (2026), Multi-adversarial metric-constrained domain generalization for road damage detection, Image and Vision Computing, 106173

[4] Danish, M. S., Khan, M. H., Munir, M. A., Sarfraz, M. S. & Ali, M. (2024), Improving single domain-generalized object detection: a focus on diversification and alignment, Proc. IEEE/CVF Conf.

Computer Vision and Pattern Recognition (CVPR)

[5] Feng, C., Zhong, Y., Gao, Y., Scott, M. R. & Huang, W. (2021), TOOD: Task-aligned one-stage object detection, Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV), pp 3490–3499

[6] Jocher, G. & Qiu, J. (2024), Ultralytics YOLO11 [software], release v8.4.192 used, https://github.com/ultralytics/ultralytics

[7] Lin, C., Tian, D., Duan, X., Zhou, J., Zhao, D. & Cao, D. (2023), DA-RDD: Toward domain adaptive road damage detection across different countries, IEEE Transactions on Intelligent Transportation Systems, 24(3), pp 3091–3103

[8] Rahman, Q. M., Sünderhauf, N. & Dayoub, F. (2019), Did you miss the sign? A false negative alarm system for traffic sign detectors, Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)

[9] Tarvainen, A. & Valpola, H. (2017), Mean teachers are better role models: weight-averaged consistency targets improve semi-supervised deep learning results, Advances in Neural Information Processing Systems (NeurIPS)

[10] Wang, J., Xu, C., Yang, W. & Yu, L. (2021), A normalized Gaussian Wasserstein distance for tiny object detection, arXiv:2110.13389

Appendix A: Plan of Action

Weeks are counted from the approved project start; dates will be fixed once the Faculty schedule is published.

Table 2: Plan of action

Weeks	Activity	Responsible
1–3	Literature study; read closest papers in full; write and commit the evaluation protocol; download and prepare RDD2022	All
4–6	Shared YOLO11 baseline; same-country vs unseen-country gap; timing pilot; scope decision	All
7–8	Method implementation; C's label check (go/no-go gate)	A, B, C
9–22	Development on inner validation countries; screened tuning; rolling final runs; global freeze in week 22	A, B, C
23–27	Final runs completed; held-out countries scored once; analysis and bootstrap	All
28–30	Prototype web application (detector, scores, miss heat map)	All
31–34	Final report, one chapter per member study; presentation and viva	All